

DYNAMICAL MASS GENERATION BY COMPACT EXTRA DIMENSIONS[☆]

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We show that if there is a physical spacelike extra dimension of a circle (S^1) with small size β , the existence of fermions can cause dynamical instability of a vacuum at low energies $\ll 1/\beta$ by quantum effects. In a system consisting of gauge fields (A_μ) and fermions with minimal gauge interactions an ordinary vacuum ($\langle A_\mu \rangle = 0$) is always destabilized by fermions obeying a periodic boundary condition in the extra dimensional coordinate. In a new vacuum the gauge symmetry remains unbroken, but fermions acquire dynamical masses by quantum corrections, even if they are massless at the tree level. We also present a toy model consisting of gauge and scalar fields in which SU(2) symmetry breaks down to U(1) without the scalar fields developing non-vanishing expectation value.

In this letter we explore a new mechanism for dynamical mass generation and gauge symmetry breaking by assuming the existence of extra dimensions in spacetime. Since Kaluza and Klein [1] introduced an extra dimension in their attempt to unify electromagnetism and gravity, many proposals have been made to incorporate other interactions, and to justify the existence of extra dimensions [2,3]. A big trend in recent investigation is that extra dimensions are not merely introduced as a mathematical trick for constructing a unified theory of various interactions, but they are considered as real and physical. The observed four-dimensionality is a consequence of compactness of extra dimensions, which have so small size that they cannot be observed at low energies. It has been even argued that the compactification of extra dimensions (or dimensional reduction) can occur spontaneously as a result of equations of motion [4]. The eleven-dimensional supergravity may be a good example for this [5]. So far, however, a ground state and an associated particle mass spectrum have been examined by solving equations of motion at the classical (tree) level.

It is obvious that if extra dimensions are physical, quantum effects in extra dimensions have to be taken into account [6]. A novel feature of our results is that

fermions, which may be massless at the tree level, acquire dynamical masses through quantum corrections at low energies. We will, furthermore, argue that in general situations, quantum effects could naturally induce gauge symmetry breaking.

To elucidate the problem, let us take a simple example. Assume that spacetime topology is given by $M = M^d \times S^1$, where M^d is a d -dimensional Minkowski space and S^1 , a spacelike extra dimension, is a circle with a circumference β . At energies much smaller than β^{-1} effective spacetime is given by M^d . We consider a system consisting of SU(N) gauge fields (A_μ) and massless fermions (ψ) with minimal gauge interactions. Gravity is neglected. (Note that A_μ are matter gauge fields, but not a part of a metric $g_{\mu\nu}$, which is taken to be flat in this note.)

Since S^1 is not a simply-connected space, boundary conditions must be specified for the theory to be defined. Let x_m and y be coordinates of M^d and S^1 , respectively. Physical observables must be single-valued on $M^d \times S^1$. We impose a periodic boundary condition (b.c.) on $A_\mu^{\pm 1}$ [i.e., $A_\mu(x, y + \beta) = A_\mu(x, y)$], but impose a more general condition on ψ [3,8];

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^{†1} In general one can impose a twisted boundary condition on A_μ and ψ [3,7]. See also discussions about eqs. (15) and (16).

$$\psi(x, y + \beta) = e^{i\delta} \psi(x, y). \quad (1)$$

This is permissible, because observables always appear in a bilinear form $\bar{\psi}\psi$.

The periodic b.c. for ψ , namely $\delta = 0$, is natural in the following sense. If an extra dimensional manifold has topology of $S^n (n \geq 2)$ so that it is simply connected, all fields should be single-valued on the manifold. $\delta = 0$ corresponds to the single-valuedness of ψ .

Recalling results in finite temperature ($T \neq 0$) field theories, we can realize the intriguing nature of the model under consideration. $T \neq 0$ field theories are defined on a manifold $R^3 \times S^1$ by imposing an anti-periodic b.c. on the imaginary time axis (S^1) on fermions. Our model with $\delta = \pi$ is related by analytic continuation to a $T \neq 0$ $SU(N)$ gauge theory, in which, as is well known, longitudinal components of gauge fields, corresponding to static electric fields, become massive [9,10];

$$m^2 = \frac{1}{3}g^2 T^2 (N + \frac{1}{2}N_F). \quad (2)$$

Here g is a coupling constant and N_F is a generation number of fermions belonging to the fundamental representation of the group. Both gauge boson and fermion loops yield positive contributions to $(\text{mass})^2$, in accordance with non-vanishing plasma at $T \neq 0$.

What happens if fermions obey a periodic b.c.? By simple calculations we see that masses of $A_{\text{longitudinal}}$, or y -component A_y , in our case, is given by

$$m^2 = \frac{1}{3}g^2 T^2 (N - N_F). \quad (3)$$

The sign of fermion contributions is reversed, and their magnitude is doubled. In our case $T = \beta^{-1}$. If $N_F > N$, (3) represents instability of a vacuum.

What is a true vacuum? What kind of field configuration is realized in the vacuum? Eq. (3) suggests $\langle A_y \rangle \neq 0$. If so, is the $SU(N)$ gauge symmetry spontaneously broken at low energies for small δ ?

We show below that the ordinary vacuum ($\langle A_y \rangle = 0$) is always destabilized by fermions obeying a periodic b.c. (even if $N_F = 1$). The gauge symmetry, however, remains unbroken, but fermions acquire dynamical masses at low energies.

To see it, we evaluate one loop effective potential V_{eff} for constant A_y . Calculations for $\delta = \pi$ have been found in ref. [9]. We first diagonalize A_y ;

$$\langle A_y \rangle = \begin{bmatrix} \phi_1 & & & \\ & \ddots & & \\ & & \ddots & \\ & & & \phi_N \end{bmatrix}, \quad (4)$$

$$\sum_{j=1}^N \phi_j = 0. \quad (5)$$

In the case of $M^3 \times S^1$, we have in the background gauge ^{#2}

$$V_{\text{eff}} = -\frac{2}{\beta^4 \pi^2} \sum_j \sum_k \sum_{n=1}^{\infty} \frac{1}{n^4} \cos n(\theta_j - \theta_k) \\ + N_F \frac{4}{\beta^4 \pi^2} \sum_j \sum_{n=1}^{\infty} \frac{1}{n^4} \cos n(\theta_j - \delta), \quad (6)$$

$$\theta_j = g\beta\phi_j, \quad \sum_j \theta_j = 0.$$

Here we have retained only θ_j -dependent terms. The first term comes from gauge field and ghost loops, while the second term is a contribution from four component Dirac fermions in the fundamental representation. Eq. (6) reproduces (2) and (3), expanded in a power series of θ_j .

V_{eff} is periodic in θ_j 's. The periodicity is related to gauge degrees of freedom. Let us make a gauge transformation described by a transformation potential

$$\Omega(x, y) = \exp \left[\frac{2\pi i y}{\beta} \begin{bmatrix} n_1 & & & \\ & \ddots & & \\ & & \ddots & \\ & & & n_N \end{bmatrix} \right],$$

$$\sum n_j = 0, \quad n_j: \text{integer}. \quad (7)$$

Note that Ω must be periodic in our case. Under the gauge transformation

$$\theta_j \rightarrow \theta_j + 2\pi n_j, \quad (8)$$

which explains the periodicity of V_{eff} .

^{#2} Useful formulae are

$$\frac{1}{\pi^4} \sum_{n=1}^{\infty} \frac{\cos n\pi x}{n^4} = \frac{1}{90} - \frac{1}{48} x^2 (x-2)^2, \quad \text{for } 0 \leq x \leq 2,$$

$$\sum_{n=-\infty}^{\infty} \frac{1 + \alpha(2\pi n - \theta)}{y^2 + (2\pi n - \theta)^2} = \frac{y^{-1} \sinh y - \alpha \sin \theta}{2(\cosh y - \cos \theta)}.$$

In M^4 a constant $A_y \neq 0$ has no physical meaning, since it is gauge equivalent to $A_y = 0$. In $M^3 \times S^1$ it is physically meaningful [9,11]. With (8) in mind we define

$$\bar{\theta}_j = \theta_j - 2\pi[\theta_j/2\pi], \quad 0 \leq \bar{\theta}_j < 2\pi$$

$$\sum \bar{\theta}_j = 2\pi g \quad (g = 0, 1, \dots, N-1). \quad (9)$$

$\bar{\theta}_j$ is gauge invariant. Indeed it is related to a gauge invariant eigenvalue $e^{i\theta_j}$ of an operator $P \exp[ig \int_0^\beta dy \times A_y(x, y)]$ [9]. At the classical level the potential is θ_j -independent. This degeneracy is lifted by quantum corrections.

An absolute minimum of V_{eff} , (6), is easily found for $SU(2)$. In this case $\theta_1 = -\theta_2 = \theta$. The absolute minimum is given by $\bar{\theta} = \theta - 2\pi[\theta/2\pi] = \pi$ for $|\delta| < \pi/2$ and $\bar{\theta} = 0$ for $\pi/2 < |\delta| \leq \pi$. For $\delta = \pm\pi/2$ these two are degenerate. The result is independent of $N_F (\geq 1)$.

For $SU(N)$ ($N \geq 3$) complete evaluation of V_{eff} is difficult. Let us take $SU(5)$ as an example. In general $\bar{\theta}_j$'s can take five different values. We look for an absolute minimum of V_{eff} by making an Ansatz that at the minimum $\bar{\theta}_j$'s take at most two different values. As we will see below, configurations satisfying this condition include $SU(5)$, $SU(4) \times U(1)$, and $SU(3) \times SU(2) \times U(1)$ symmetric states. The result is again independent of $N_F (\geq 1)$. The absolute minimum is found at all $\bar{\theta}_j$'s = $\bar{\theta}$. $\bar{\theta} = \frac{2}{5}q\pi$ for $(-\frac{6}{5} + \frac{2}{5}q)\pi < \delta < (-\frac{4}{5} + \frac{2}{5}q)\pi \pmod{2\pi}$, $q = 0, 1, 2, 3, 4$. At $\delta = \frac{2}{5}q\pi \pmod{2\pi}$ there are two degenerate minima^{#3}.

For $1 \leq N_F \leq N$, the ordinary vacuum is a local, but not a global minimum of the potential.

Before discussing implications of the new vacuum, we note trivial generalization of (6). The second term in (6) is for fermions belonging to the fundamental representation of the group. If fermions are belonging to, for instance, **10** representations of $SU(5)$, we have

$$N_F \frac{4}{\beta^4 \pi^2} \sum_{j < k} \sum \frac{1}{n^4} \cos n(\theta_j + \theta_k - \delta). \quad (10)$$

For $\delta = 0$, an absolute minimum is found at all $\bar{\theta}_j$'s = $\bar{\theta} = \frac{2}{5}\pi$ or $\frac{4}{5}\pi$, as opposed to $\bar{\theta}_j = \bar{\theta} = \frac{4}{5}\pi$ or $\frac{6}{5}\pi$ for fermions $\in 5$. If there are the same number of **5** and **10**, **10**-fermions strongly dominate to determine a true vacuum.

^{#3} Representatives of $\bar{\theta}_j = \bar{\theta} = \frac{4}{5}\pi$, for instance, are $\theta_j = \frac{4}{5}\pi \times (1, 1, 1, 1, -4)$ and $\frac{2}{5}\pi \times (2, 2, 2, -3, -3)$.

The expression (6) is for $M^3 \times S^1$. If we consider $M^d \times S^1$, we would have $(\beta n)^{-d-1}$ behavior in place of $(\beta n)^{-4}$. We see that the instability of the ordinary vacuum is a general feature irrespective of d .

Properties of the new vacuum become clear by examining a particle mass spectrum at low energies ($\ll \beta^{-1}$). The y -component of gauge fields, A_y , becomes a Higgs scalar field in the adjoint representation, acquiring a mass of $O(g/\beta)$ analogously to (2). To see masses of other components of gauge fields, A_m , we expand A_m in a Fourier series in y :

$$(A_m)_{jk} = \sum_n (A_m^{[n]})_{jk} \exp(2\pi i n y / \beta),$$

$$(A_m^{[-n]})_{jk} = (A_m^{[n]})_{jk}^*, \quad n: \text{integer}. \quad (11)$$

From $\mathcal{L} \sim -\text{tr } F_{my} F^{my}$ we find, with $\langle A_y \rangle \neq 0$ in mind,

$$\int_0^\beta \frac{dy}{\beta} \mathcal{L} \sim -\left(\frac{2\pi}{\beta}\right)^2 \sum_j \sum_k \sum_n [n - (\theta_j - \theta_k)/2\pi]^2$$

$$\times |(A_m^{[n]})_{jk}|^2. \quad (12)$$

The mass spectrum is given by

$$(2\pi/\beta)|n - (\theta_j - \theta_k)/2\pi| \quad (n: \text{integer}). \quad (13)$$

If all $\bar{\theta}_j$'s = $\bar{\theta}$, all $SU(N)$ components of A_m have the same mass spectrum given by $(2\pi n/\beta)$ ($n: \text{integer}$). Among various mass eigenstates are massless gauge fields, i.e., the gauge symmetry is not broken. If $\bar{\theta}_j \neq \bar{\theta}_k$ for some j and k , the symmetry is spontaneously broken. For instance, $(\bar{\theta}_a, \bar{\theta}_a, \bar{\theta}_a, \bar{\theta}_b, \bar{\theta}_b)$ and $(\bar{\theta}_a, \bar{\theta}_a, \bar{\theta}_a, \bar{\theta}_a, \bar{\theta}_b)$ ($\bar{\theta}_a \neq \bar{\theta}_b$) in $SU(5)$ corresponds to residual $SU(3) \times SU(2) \times U(1)$ and $SU(4) \times U(1)$ symmetry, respectively. In our case the true vacuum, in which all $\bar{\theta}_j$'s = $\bar{\theta}$, is $SU(N)$ symmetric.

From the constraint in (9) we see that there are N different $SU(N)$ symmetric states given by all $\bar{\theta}_j$'s = $\bar{\theta} = 2\pi q/N$ ($q = 0, 1, 2, \dots, N-1$). It is important to notice that physics remains $SU(N)$ symmetric for special values of non-vanishing $\bar{\theta}$ or $\langle A_y \rangle$, only when an infinite number of component fields $(A_m^{[n]}, n: \text{integer})$ are taken into account.

The N $SU(N)$ symmetric states are distinguished by the fermion mass spectrum at low energies. Noting that the term $\bar{\psi} \gamma^\nu (i\partial_y + gA_y) \psi$ in the lagrangian yields

masses in lower dimensions, we see that fermions in the fundamental representation have masses given by

$$(2\pi/\beta)|n + (\bar{\theta} - \delta)/2\pi| \quad n: \text{integer}. \quad (14)$$

That is, if $\bar{\theta} - \delta \neq 0 \pmod{2\pi}$, even $n = 0$ components, which are massless at the tree level for $\delta = 0$, acquire huge dynamical masses of $O(2\pi/\beta)$. In SU(5) we found $\frac{4}{5}\pi \leq \bar{\theta} - \delta \leq \frac{6}{5}\pi$ in the true vacuum, so there are no light fermions $\in 5$.

In the SU(2) case (14) implies that even if one imposes a periodic b.c. on fermions, the final mass spectrum is the same as that obtained with an anti-periodic b.c. imposed. In the SU(5) case the mass spectrum is periodic in δ with periodicity $\frac{2}{5}\pi$. This is due to the gauge invariance. To see it, let us make an SU(N) gauge transformation

$$\Omega(y) = \exp \left[\frac{2\pi i y}{\beta N} \begin{pmatrix} 1 & & & \\ & 1 & & \\ & & \ddots & \\ & & & 1 & \\ & & & & -N+1 \end{pmatrix} \right] \\ \equiv \exp(ig_y \Lambda_0). \quad (15)$$

New gauge and fermion fields are given by $A'_\mu = \Omega(A_\mu - \delta_{\mu y} \Lambda_0) \Omega^\dagger$ and $\psi' = \Omega \psi$. A' still satisfies a periodic b.c., but ψ' obeys, instead of (1),

$$\psi'(x, y + \beta) = \exp[i(\delta + 2\pi/N)] \psi'(x, y). \quad (16)$$

Therefore physics depends on δ only in mod $2\pi/N$.

Another characteristic of the new vacuum is that there is non-vanishing fermion condensate. It is easy to see

$$\langle \bar{\psi} \gamma_y \psi \rangle = \sum_j \beta \frac{\partial}{\partial \theta_j} V_{\text{eff}}[\text{fermion } \psi \text{ loop}] \\ = - \frac{4}{\beta^3 \pi^2} \sum_j \sum_{n=1}^{\infty} \frac{1}{n^3} \sin n(\theta_j - \delta). \quad (17)$$

Note that (17) is a SU(N) singlet.

To summarize, if fermions obey a periodic b.c., the ordinary vacuum ($\langle A_y \rangle = 0$) is always destabilized by quantum effects. In the new vacuum ($\langle A_y \rangle \neq 0$), the SU(N) symmetry remains unbroken, but fermions acquire huge dynamical masses at low energies. Quantum corrections are crucial to determine the mass spectrum.

So far we have discussed the gauge field-fermion

system. If we consider complex scalar fields instead of fermions, SU(2) gauge symmetry can be dynamically broken by the mechanism discussed above. Suppose that a scalar field (ψ) potential is such that at a minimum $\langle \psi \rangle = 0$. V_{eff} for gauge fields has almost the same form as (6), except that a scalar field contribution is $-\frac{1}{2} \times$ (fermion contribution). One can see that if $1 - \frac{1}{3}\sqrt{3} < |\delta/\pi| < \frac{1}{3}\sqrt{3}$, the SU(2) symmetry breaks down to U(1) for sufficiently large N_s , where N_s is the number of doublet scalar fields. For $\delta = \pi/2$, for instance, a global minimum is given by $\theta_1 = -\theta_2 = \pm\pi/2$ if $N_s > 16$.

We close our discussions by raising several questions. First note that for $\delta = 0$ (i) the gauge symmetry is not broken, (ii) fermions acquire huge dynamical masses, i.e., there are no light fermions at low energies, and (iii) the theory is left-right symmetric from the beginning to the end. All these features are undesirable in constructing realistic higher dimensional theories. Are these problems cured by considering an extra dimensional space like S^n ($n \geq 2$)? We considered a simple gauge field-fermion system. If a theory is supersymmetric and supersymmetry is unbroken, the effects discussed in the paper are exactly cancelled between fermion and boson contributions. What happens if supersymmetry is spontaneously broken? The question would be relevant in higher dimensional supergravity theories.

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Note added. After submitting the paper, I received a paper by David J. Tom [12]. He has discussed the instability of the ordinary vacuum and got the same result for massless fermions as eq. (3) in the present paper.

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